DIGITAL PAYMENTS INFRASTRUCTURE

BanglaPay

Unifying Bangladesh's Digital Payments Ecosystem

A Bank-First, MFS-Inclusive Superapp for Seamless Financial Integration

STRATEGIC OVERVIEW

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Key Metrics

Tk 17.37L

Crore Annual Transactions

28.4%

Year-over-Year Growth

CHAPTER ONE

01

Executive Overview & & Market Opportunity

Understanding BanglaPay's strategic positioning in Bangladesh's $208 billion digital payments landscape

EXECUTIVE OVERVIEW

Infrastructure-First Payment Aggregator

Superapp Aggregation Layer

BanglaPay sits atop Bangladesh's fragmented digital payments ecosystem, aggregating **multiple MFS providers, banks, and PSPs** into a single AI-powered interface without displacing incumbent providers.

National Infrastructure Leverage

Utilizes **National Payment Switch Bangladesh (NPSB)** and **BanglaQR standards** to enable real-time, cross-wallet transfers and universal merchant acceptance across platforms.

Collaborative Positioning

Ecosystem enabler, not competitor. Complements existing providers by solving user and merchant pain points through aggregation—essential for regulatory approval and partner alignment.

Tk 17.37L
Crore Transaction Volume

28.4%
YoY Growth

10,000+
Crore Daily by 2026

Market Context

Explosive Growth Growth

Bangladesh's MFS market reached **Tk 17.37 lakh crore in 2024**, with daily transactions peaking at Tk 1.64 lakh crore and projections exceeding Tk 10,000 crore daily within two years.

Fragmentation Crisis Crisis

Users juggle **3-5 wallet apps** with different UIs, balances, and transfer rules. Merchants display multiple QR codes, creating confusion and reducing payment acceptance.

Digital Readiness Readiness

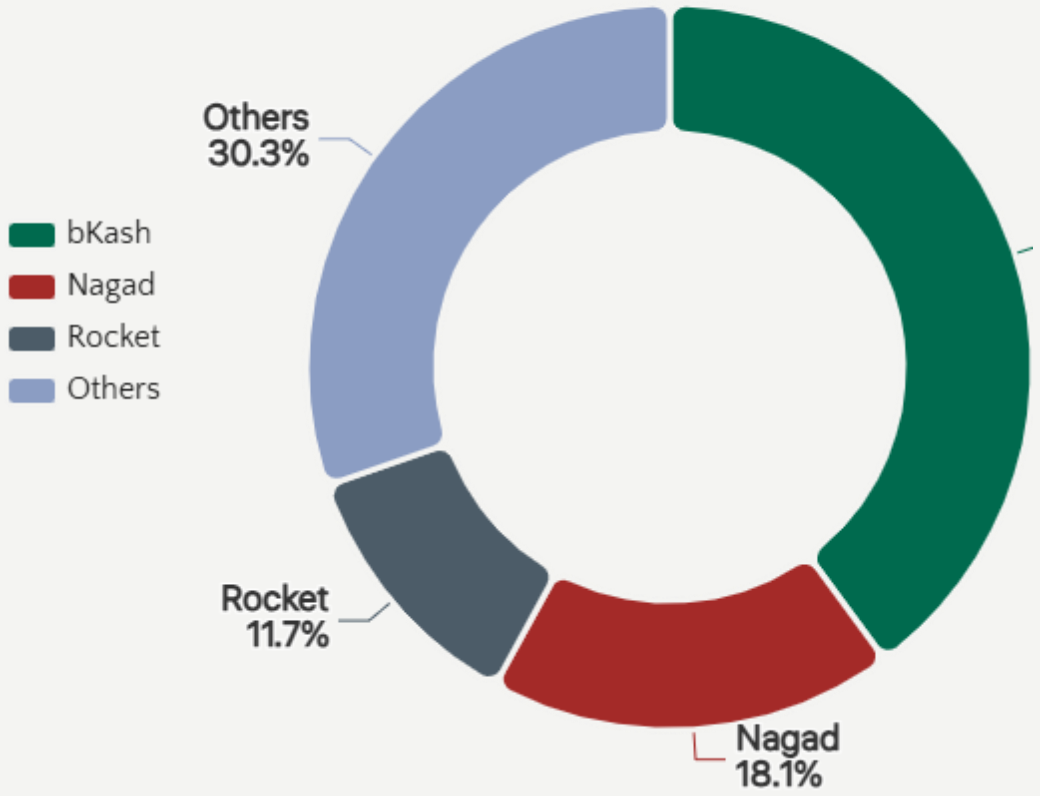
63% smartphone penetration and **45% internet access** create foundation for digital financial inclusion, yet adoption remains stymied by ecosystem frictions.

MARKET ANALYSIS

Fragmented Giants: MFS Market Dominance

Despite rapid growth, Bangladesh's digital payments ecosystem remains highly fragmented

MFS Market Share Distribution



39.9%
bKash Market Share

Market leader with **38M active users** processing **100M+ monthly transactions**. Widely used across urban and rural areas, establishing MFS as default for P2P and small merchant payments.

18.1%
Nagad Share

Rapid growth with **25M users** and **50-60M monthly transactions**. Offers bank-linked wallet option, positioning as hybrid solution between traditional MFS and banking.

11.7%
Rocket Share

Bank-backed MFS with **12M users** and **20-25M monthly transactions**. Strong urban presence through Dutch-Bangla Bank network.

30.3%
Others Combined

10+ smaller MFS operators including Upay (5-8M users combined). Fragmented further by niche players and regional providers.

The Fragmentation Problem
Problem

3-5
Wallet Apps per User

4+
QR Codes per Merchant

1-2%
Cash-Out Fees

CHAPTER TWO

02

Digital Payments Ecosystem Analysis

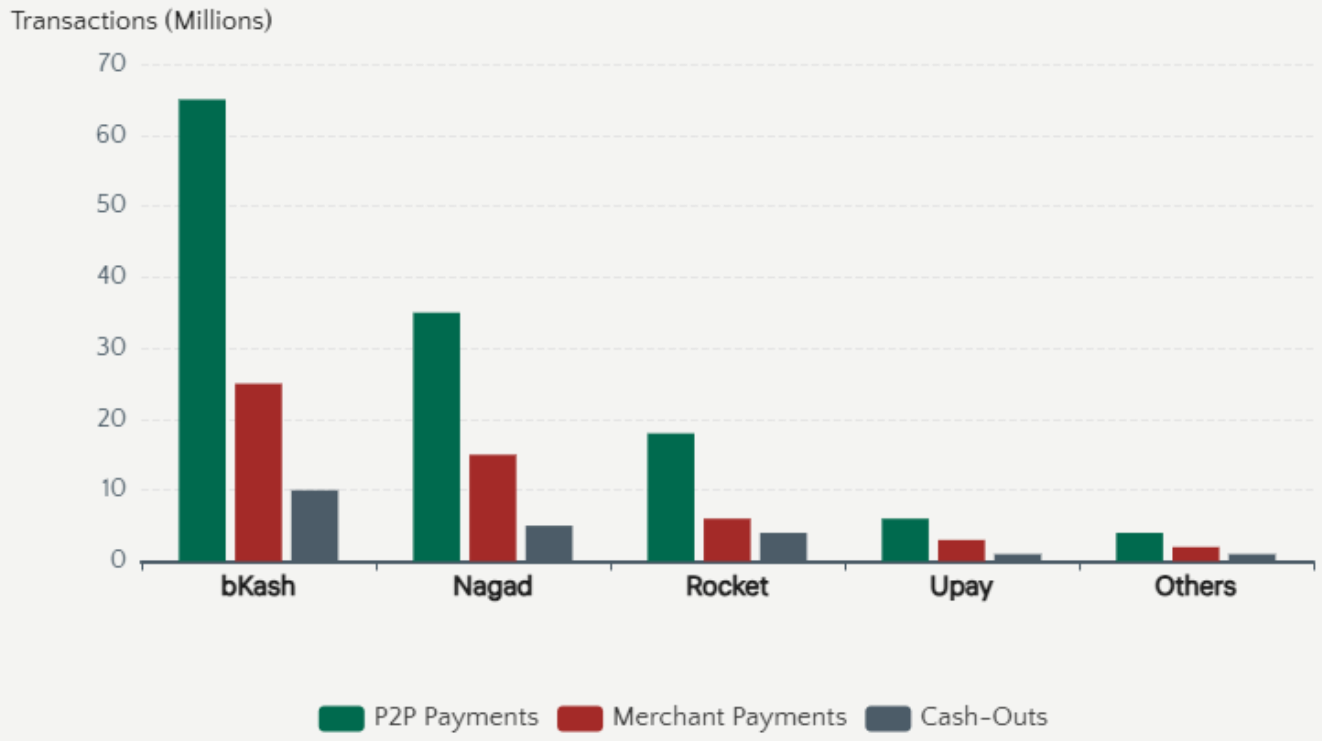
Deep dive into Bangladesh's payment infrastructure, MFS impact, and central bank initiatives

ECOSYSTEM ANALYSIS

MFS Revolution & Transaction Dominance

Mobile Financial Services have become the default for everyday digital transactions

Monthly Transaction Volumes (Millions)



Key Market Insights

Cashless Culture

MFS created **cashless culture for small payments**, especially in urban areas. Real-time P2P transfers reduced dependency on physical cash.

Micro-Merchant Enablement

Enabled **micro-merchants and SMEs** to accept payments digitally without complex POS systems, accelerating financial inclusion.

Closed-Loop Fragmentation

Fragmentation

Each MFS acts as **closed loop**, requiring cash-outs to banks and adding cost, friction. Interoperability remains limited.

Total MFS Users **~80M**

BK **bKash**
Market Leader

38M	100M+	~55%
Active Users	Monthly Txns	Market Share

N **Nagad**
Fast Follower

25M	~55M	~30%
Active Users	Monthly Txns	Market Share

R **Rocket + Others**
Niche Players

17M+	~30M	~15%
Combined Users	Monthly Txns	Market Share

REGULATORY INFRASTRUCTURE

Bangladesh Bank's Digital Infrastructure Push

Central bank initiatives driving the transition to a cashless, interoperable digital payments ecosystem

Binimoy

Early national payment switch attempt to interconnect banks for cross-bank digital payments. Faced limited adoption and no real-time settlement.

Limited Success

NPSB

Current interbank switch enabling bank-to-bank transactions, POS, and ATM interoperability. Operational since 2019, mandated since Nov 2025.

Active Rail

BanglaQR

Brand-neutral QR standard for merchant acceptance across banks and MFS. "One QR for all" goal, but limited adoption.

Adoption Gap

Digital Banks

Digital-only banking licenses issued to expand financial services. Integration with NPSB/IIPS ongoing. Sets stage for bank-first apps.

Future-Ready

IIPS

Next-gen real-time payment system connecting banks, MFS, and digital banks. Promises instant settlement and true interoperability.

Upcoming

Infrastructure Maturity Timeline

- **2019** — **NPSB Launch:** Interbank switch operational
- **2021-23** — **BanglaQR Rollout:** Standardized QR deployment
- **2025** — **NPSB Mandated:** Interoperability mandatory
- **2025-26** — **IIPS Launch:** Real-time payment system
- **2025-27** — **Digital Banks:** Full-scale operations

BanglaPay Alignment

BanglaPay is designed as the **execution layer for this infrastructure evolution**, bridging gaps between NPSB, BanglaQR, and upcoming IIPS while maintaining MFS inclusivity.

- Leverages NPSB for bank-first routing today
- Adopts BanglaQR as universal merchant standard
- Future-proofs for IIPS real-time rails
- Supports digital bank ecosystem growth

CHAPTER THREE

03

Systemic Challenges & Pain Points

Analyzing transaction costs, interoperability barriers, and ecosystem friction limiting digital financial inclusion

SYSTEMIC CHALLENGES

Payment Rails & Interoperability Gap

Three competing standards with varying adoption and cost structures

BanglaQR

- Brand-neutral, standardized QR
- Low settlement fees (~0.1-0.2%)
- Limited merchant adoption
- Consumer awareness gap

Status: Underutilized

NPSB

- Low-cost bank rails (<0.1%)
- Regulated, standardized processing
- Limited consumer adoption
- Partial merchant integration

Status: Active but Underused

IIPS

- Real-time settlement
- True interoperability
- Risk: Adoption barriers
- Still in rollout phase

Status: In Development

Transaction Cost Comparison

Bank-to-Bank NPSB Rails	**<0.1%**
MFS-to-MFS Closed Loop	**1-2%**
Cross-Platform Bank ↔ MFS	**0.2-1%**

Key Insight: Bank rails reduce costs by **60-90%** versus MFS cash-out, but adoption limited by ecosystem inertia and convenience factors.

Why BanglaPay Works

BanglaPay bridges these fragmented rails through **intelligent routing**, automatically selecting the lowest-cost path for each transaction while maintaining seamless user experience.

1. Bank-first routing via NPSB/IIPS
2. MFS inclusion for compatibility
3. BanglaQR standardization
4. Future-proof for IIPS migration

60-90%

Cost Reduction Potential

ECOSYSTEM FRICTION

Consumer & Merchant Pain Points Matrix

Daily friction points limiting digital financial inclusion and ecosystem growth

Consumer Pain Points Points

01 Fragmented Apps

Consumers need **multiple apps** for bank and MFS payments. Juggling 3-5 wallets creates cognitive overload and app proliferation fatigue.

Daily High Impact

02 Checkout Confusion

Different QR codes at merchants create uncertainty. Consumers abandon purchases when preferred wallets or QR codes aren't accepted.

Daily High Impact

03 High Transaction Costs

MFS cash-out or cross-platform transfers cost **1-2% per transaction**. For frequent users, this amounts to significant monthly expenses.

Daily Economic Impact

04 Trust & Security Security

Multiple wallets, unknown fees, and lack of real-time tracking reduce trust. Security concerns across different platforms create hesitation.

Ongoing Trust Impact

Merchant Pain Points Points

01 Multiple Acceptance

Need to display **multiple QR codes** or manage several POS systems. Rahim's tea stall may display 4 different QR codes, confusing customers.

Daily Operational Impact

02 High Cash-Out Fees

MFS cash-out fees of **1-2%** increase operational costs. Micro-merchants operate on thin margins, making these fees significant.

Daily Economic Impact

03 Reconciliation Complexity Complexity

Tracking transactions across different rails and apps is time-consuming. Manual reconciliation creates errors and cash flow uncertainty.

Daily Time Impact

04 Brand Hijacking

Proprietary MFS QR codes overshadow BanglaQR, creating fragmented customer experience. Limits merchant flexibility.

Ongoing Strategic Impact

CHAPTER FOUR

04

BanglaPay Solution Architecture

Comprehensive overview of features, technical design, and strategic pillars driving unified payments

SOLUTION ARCHITECTURE

Core Proposition & Feature Pillars

Six strategic pillars unifying Bangladesh's fragmented digital payments ecosystem

Unified Rails

Bank-first routing via NPSB and upcoming IIPS for cost efficiency, with MFS inclusion for ecosystem compatibility and user flexibility.

- ✓ Auto-selects lowest-cost rail
- ✓ Transparent fee display
- ✓ Future-proof for IIPS

BanglaQR Standard

One QR for all merchants and consumers. Dynamic QR generation with real-time routing via NPSB infrastructure for universal acceptance.

- ✓ Universal merchant acceptance
- ✓ Cross-platform compatibility
- ✓ POS/API integration ready

Merchant-Friendly

Zero cash-out fees with direct bank settlement. Comprehensive dashboards for payment tracking, automated reconciliation, and customer analytics.

- ✓ Instant bank settlement
- ✓ Automated reconciliation
- ✓ Customer analytics

Intelligent Routing Routing

AI-powered system **auto-selects lowest-cost rail** per transaction. Optimizes for cost, speed, and reliability while maintaining transparent user experience.

- ✓ Cost optimization algorithms
- ✓ Real-time route selection
- ✓ Failure fallback mechanisms

Walletless Architecture

Consumers can **pay directly from bank accounts**, eliminating need to preload wallets. Reduces friction and removes cash-out costs entirely.

- ✓ Direct bank payments
- ✓ No preloading required
- ✓ Multi-account aggregation

Future-Proof Design Design

Compatible with **digital banks and IIPS rails**. Scales with infrastructure evolution and supports hybrid transactions across all payment types.

- ✓ IIPS integration ready
- ✓ Digital bank support
- ✓ Scalable architecture

FEATURE DEEP DIVE

Five Core Features in Action

Comprehensive feature set delivering seamless, secure, and intelligent payment experiences

Multi-Account Dashboard

Unified balance view across MFS wallets, bank accounts, and institutional accounts in real-time.

- Combined balances, consolidated history
- Munshi AI categorization
- Searchable, exportable data

Cross-Platform Transfers

Fast, low-cost P2P transfers between any linked accounts with transparent, upfront fee display.

- One-tap send money flow
- Automated fee calculation
- Real-time confirmation

Universal BanglaQR

Dynamic QR codes for payments from any wallet or bank app with real-time routing.

- One QR for all apps
- Real-time NPSB routing
- Merchant portal & analytics

Munshi AI Platform Platform

Smart insights & recommendations using machine learning for spending optimization.

- Transaction categorization
- Bill reminders & auto-pay
- Fraud anomaly detection

Security & Compliance Compliance

Biometric and PIN authentication with ML-driven fraud detection and end-to-end encryption.

- Device native biometrics
- TLS 1.3 & AES-256 encryption
- Bangladesh Bank compliant

User Experience Flow

1. **Scan** — BanglaQR code
2. **Enter** — Amount
3. **Authorize** — Biometric/PIN
4. **Confirm** — Instant settlement

Merchant Benefits

- One QR for all customers
- Zero cash-out fees
- Instant bank settlement
- Consolidated analytics
- Reduced reconciliation time

CHAPTER FIVE

05

Impact, Strategy & Roadmap

Cost reduction analysis, ecosystem positioning, competitive advantages, and phased implementation plan

IMPACT ANALYSIS

Cost Reduction & Zero-MDR Impact

Quantifying BanglaPay's economic benefits for consumers and merchants

Monthly Cost Comparison: 100 Transactions × 1,000 BDT

Consumer Benefits

90%

Transaction Cost Reduction

Zero

Cash-Out Fees

- Single app experience
- Seamless BanglaQR payments
- Direct bank payments
- True interoperability

Zero-MDR Strategy Framework

1. **Centralized MDR Absorption**
   MDR is **zero for merchants**, especially micro-merchants. System costs absorbed by banks and PSPs through ecosystem optimization.

2. **Intelligent Acquirer Rotation**
   No single bank carries cost burden. System **rotates acquirers** fairly based on capacity and coverage.

3. **Volume-Driven Recovery**
   Increased digital adoption drives **higher float, retention, cross-selling**, subsidizing Zero-MDR model.

GLOBAL LESSONS

UPI Analogy: India's Payments Revolution

Parallels between India's UPI success and Bangladesh's digital payments opportunity

Pre-UPI India

Wallet Dominance

Paytm and other wallets dominated, but required cash-in/cash-out with associated fees and inefficiencies

Fragmented Ecosystem

Multiple apps, closed-loop systems, and limited interoperability created user friction

High Merchant Costs

MDR fees of 1-2% limited digital adoption, especially among micro-merchants

Post-UPI India

Bank-First Model

Transactions became instant and low-cost, promoting bank rails while keeping wallets optional

Zero MDR

Merchants benefited from zero cash-out costs, simplifying reconciliation and encouraging adoption

Explosive Growth

Digital adoption skyrocketed across urban and rural India, with apps like PhonePe and Google Pay leading

Bangladesh Parallels

- **High mobile adoption** but fragmented MFS ecosystem
- **Fragmented merchant acceptance** mirrors India pre-UPI
- **Closed-loop wallets** with high cash-out fees
- **NPSB infrastructure** underutilized like early NPCI

80M+

MFS Users Ready for Unified Solution

BanglaPay Opportunity

Bank-first, interoperable apps like BanglaPay can **replicate UPI's success** by addressing similar conditions and learning from India's implementation.

1. **Infrastructure Alignment**
   Leverage NPSB today, IIPS tomorrow—similar to UPI's NPCI foundation
2. **Zero-MDR Flywheel**
   Follow UPI's zero-MDR model to accelerate merchant adoption
3. **Ecosystem Partnership**
   Collaborative positioning as enabler, not competitor
4. **AI-Powered UX**

STRATEGIC POSITIONING

Ecosystem Positioning & Competitive Advantage

BanglaPay as infrastructure-level glue, not consumer-facing competitor

Ecosystem Enabler, Not Competitor

BanglaPay is intentionally designed as **infrastructure-level glue**, not a consumer-facing competitor. It does not challenge bKash, Nagad, Rocket, or bank apps—it strengthens them.

Aggregates
Bank accounts and mobile wallets into one unified interface

Standardizes
Merchant acceptance through one QR aligned with BanglaQR standards

Reduces Costs
Zero-MDR strategy enables micro-merchants to accept digital payments without charges

Unifies Rails
Makes NPSB and IIPS work frictionlessly together

Expands Volume
Increases everyone's transaction volume rather than shrinking market share

Trust & Collaboration Design

User Control

Control
Multi-wallet aggregation only with **explicit user permission**. No hidden flows—users see which app/bank handles each transaction.

Provider Respect

Respect
Banks and MFS maintain **full ownership** of their users, KYC, and balances. BanglaPay doesn't store money or replace any app.

Localized Experience

Experience
Bangla-first UX for mass-market trust. Rural-friendly interface with agent-support options and transparent fee display.

Five Competitive Advantages

1 Non-Competitive

Supports bKash/Nagad instead of replacing them. Expands banks' digital footprint without UI redesigns.

2 Universal QR

Single BanglaPay QR works for all bank apps, MFS apps, and future IIPS integrations.

3 Multi-Wallet

First in Bangladesh: all balances in one place, all transfers in one interface.

4 Zero-MDR

Fastest growth flywheel: when merchants pay zero, adoption skyrockets and QR penetration increases.

5 Infrastructure

Future-proof UX layer masking backend complexity, ensuring stable merchant acceptance.

IMPLEMENTATION STRATEGY

Phased Roadmap & Risk Mitigation

Three-phase rollout strategy with comprehensive risk management framework

1 Phase 1: Foundation (0-6 Months)

50K
Users

1,000
Merchants

- NPSB transfers + BanglaQR
- Closed beta launch
- Multi-wallet aggregation
- Intense user feedback collection

2 Phase 2: Growth (6-18 Months)

1M
Users

20K
Merchants

- Public launch
- Munshi AI features
- Merchant tools + loyalty
- Referral incentives

3 Phase 3: Scale (18-36 Months)

5M+
Users

100K
Merchants

- Advanced AI (credit scoring)
- Open APIs for fintechs
- IIPS integration
- Regional expansion

IMPLEMENTATION STRATEGY

Risk Mitigation

Three-phase rollout strategy with comprehensive risk management framework

Risk Assessment Matrix

Partner Resistance MEDIUM

MFS providers may restrict API access or delay partnerships due to competitive concerns.

Mitigation: Start with willing partners, demonstrate value through pilot metrics, engage regulators early for ecosystem benefits.

User Adoption MEDIUM

Users may be reluctant to switch from familiar MFS apps or adopt another aggregator platform.

Mitigation: Invest in simple UX superior to incumbents, educational campaigns, incentives and referral programs, agent-assisted onboarding.

Regulatory Barriers LOW

Licensing requirements, compliance costs, and regulatory oversight may delay launch.

Mitigation: Proactive communication with Bangladesh Bank, adopt non-custodial model reducing regulatory burden, align with national infrastructure goals.

Fraud & Security MEDIUM

Increased attack surface with aggregated accounts creates new vulnerabilities.

Mitigation: Invest in ML-driven fraud detection, biometrics, user education on security best practices, regular security audits.

CATALYZING BANGLADESH'S DIGITAL FUTURE

Unifying Payments, Empowering Millions

BanglaPay offers a pioneering, integrative solution to fragmentation challenges by leveraging national infrastructure, AI-powered insights, and a cooperative trust-first approach. By focusing on collaboration over competition, BanglaPay unlocks untapped market potential, increases transaction volumes, and empowers underserved segments.

60-90%

Cost Reduction for Merchants

5M+

Users by Phase 3

100K+

Merchants by 2027

"Paving the way for a truly inclusive, efficient, and modern digital payments ecosystem in Bangladesh"